Summer 1982

Problem 1 (Sp78, Sp82, Su82, Fa90). Determine the Jordan Canonical Form of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 4 \end{pmatrix}$$

Problem 2 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1+x^{2n}}$, where n is a positive integer.

Problem 3 (Su82, Sp95). Let K be a nonempty compact set in a metric space with distance function d. Suppose that $\varphi\colon K \to K$ satisfies

$$d(\varphi(x), \varphi(y)) < d(x, y)$$

for all $x \neq y$ in K. Show there exists precisely one point $x \in K$ such that $x = \varphi(x)$.

Problem 4 (Su82). Let G be a group with generators a and b satisfying

$$a^{-1}b^2a = b^3 \qquad b^{-1}a^2b = a^3$$

Is G trivial?

Problem 5 (Su82). Let $0 < a_0 \leqslant a_1 \leqslant \cdots \leqslant a_n$. Prove that the equation

$$a_0z^n + a_1z^{n-1} + \cdots + a_n = 0$$

has no roots in the disc $|z| < 1$.

Problem 6 (Sp77, Su82, Fa18, Fa21). Suppose f is a differentiable real valued function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$ and $f(0) = 0$. Prove that $f(x) > 0$ for all positive x.

Problem 7 (Su82). Let $A \in M_{3\times 3}$ be the diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Find the determinant of the linear transformation $T : M_{3\times 3} \to M_{3\times 3}$ defined by $T(X) = (1/2)(AX + XA)$.

Problem 8 (Su82). Let n be a positive integer.

1. Show that the binomial coefficient $c_n = \binom{2n}{n}$ is even.

2. Prove that c_n is divisible by 4 if and only if n is not a power of 2.

Problem 9 (Su82). Determine the complex numbers z for which the power series

$$\sum_{n=1}^{\infty} \frac{z^n}{n^{\log n}}$$

and its term by term derivatives of all orders converge absolutely.

Problem 10 (Su82). For complex numbers $\alpha_1, \alpha_2, \ldots, \alpha_k$, prove

$$\limsup_n \left| \sum_{j=1}^{k} \alpha_j^n \right|^{1/n} = \sup_j |\alpha_j|$$

Note: See also **??**.

Problem 11 (Su82). Let $s(y)$ and $t(y)$ be real differentiable functions of y, $-\infty < y < \infty$, such that the complex function

$$f(x+iy) = e^x \left(s(y) + it(y)\right)$$

is complex analytic with $s(0) = 1$ and $t(0) = 0$. Determine $s(y)$ and $t(y)$.

Problem 12 (Su81, Su82). Show that $x^{10}+x^9+x^8+\cdots+x+1$ is irreducible over $\mathbb{Q}$. How about $x^{11} + x^{10} + \cdots + x + 1$?

Problem 13 (Su82). Let $f : [0, \pi] \to \mathbb{R}$ be continuous and such that

$$\int_0^{\pi} f(x) \sin nx \, \mathrm{d}x = 0$$

for all integers $n \geqslant 1$. Prove that $f(x)$ is identically 0.

Problem 14 (Su82). Let A be a real $n \times n$ matrix such that $\langle Ax, x\rangle \geqslant 0$ for every real vector $x \in \mathbb{R}^n$. Show that $Au = 0$ if and only if $A^{\mathsf{T}} u = 0$.

Problem 15 (Su82). Let $f(z)$ be analytic on the open unit disc $\mathbb{D} = \{z \mid |z|<1\}$. Prove that there is a sequence (z_n) in $\mathbb{D}$ such that $|z_n| \to 1$ and $(f(z_n))$ is bounded.

Problem 16 (Sp77, Su82, Sp24). A square matrix A is nilpotent if $A^k = 0$ for some positive integer k.

1. If A and B are nilpotent, is $A + B$ nilpotent?
2. Prove: If A and B are nilpotent matrices and $AB = BA$, then $A + B$ is nilpotent.
3. Prove: If A is nilpotent then $I + A$ and $I - A$ are invertible.

Problem 17 (Su82). Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ and assume that 0 is a *regular value* of f, i.e., the differential of f has rank 2 at each point of $f^{-1}(0)$. Prove that $\mathbb{R}^3 \setminus f^{-1}(0)$ is arcwise connected.

Problem 18 (Su82). Let E be the set of all continuous real valued functions $u : [0, 1] \to \mathbb{R}$ satisfying

$$|u(x) - u(y)| \leqslant |x - y| \qquad 0 \leqslant x, y \leqslant 1 \qquad u(0) = 0$$

Let $\varphi : E \to \mathbb{R}$ be defined by

$$\varphi(u) = \int_0^1 \left(u(x)^2 - u(x)\right) \mathrm{d}x$$

Show that φ achieves its maximum value at some element of E.

Problem 19 (Su81, Su82). Let V be a finite dimensional vector space over the rationals $\mathbb{Q}$ and let M be an automorphism of V such that M fixes no nonzero vector in V. Suppose that M^p is the identity map on V, where p is a prime number. Show that the dimension of V is divisible by $p - 1$.

Problem 20 (Su82). Let $M_{2\times 2}$ be the four-dimensional vector space of all 2×2 real matrices and define $f : M_{2\times 2} \to M_{2\times 2}$ by $f(X) = X^2$.

1. Show that f has a local inverse near the point $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
2. Show that f does not have a local inverse near the point $X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.